

1. At time  $t$  seconds,  $t \geq 0$ , a particle  $P$  has position vector  $\mathbf{r}$  metres with respect to a fixed origin  $O$ , where

$$\mathbf{r} = (t^3 - 8t)\mathbf{i} + \left(\frac{1}{3}t^3 - t^2 + 2t\right)\mathbf{j}$$

- (a) Find the acceleration of  $P$  when  $t = 4$

(5)

At time  $T$  seconds,  $T \geq 0$ ,  $P$  is moving in the direction of  $(2\mathbf{i} + \mathbf{j})$

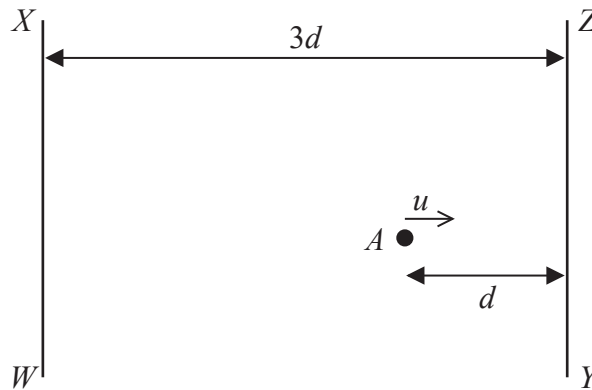
- (b) Find the value of  $T$

(3)

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2.



**Figure 1**

The point  $A$  lies on a smooth horizontal floor between two fixed smooth parallel vertical walls  $WX$  and  $YZ$ , as shown in the plan view in Figure 1.

The distance between  $WX$  and  $YZ$  is  $3d$ .

The distance of  $A$  from  $YZ$  is  $d$ .

A particle is projected from  $A$  along the floor with speed  $u$  towards  $YZ$  in a direction perpendicular to  $YZ$ .

The coefficient of restitution between the particle and each wall is  $\frac{2}{3}$

The time taken for the particle to move from  $A$ , bounce off each wall once and return to  $A$  for the **first** time is  $T_1$

(a) Find  $T_1$  in terms of  $d$  and  $u$ .

**(5)**

The ball returns to  $A$  for the first time after bouncing off each wall once.

The further time taken for the particle to move from  $A$ , bounce off each wall once and return to  $A$  for the **second** time is  $T_2$

(b) Find  $T_2$  in terms of  $d$  and  $u$ .

**(1)**



3. A particle  $P$  of mass  $0.5 \text{ kg}$  is moving with velocity  $\lambda(\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$  when  $P$  receives an impulse of magnitude  $\sqrt{\frac{5}{2}} \text{ N s}$

Immediately after  $P$  receives the impulse, the velocity of  $P$  is  $4\mathbf{i} \text{ m s}^{-1}$   
Given that  $\lambda$  is a constant, find the two possible values of  $\lambda$

(6)

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4. A truck of mass  $900\text{ kg}$  is moving along a straight horizontal road with the engine of the truck working at a constant rate of  $P$  watts. The resistance to the motion of the truck is modelled as a constant force of magnitude  $R$  newtons.

At the instant when the speed of the truck is  $15\text{ m s}^{-1}$ , the deceleration of the truck is  $0.2\text{ m s}^{-2}$

Later the same truck is moving down a straight road inclined at an angle  $\theta$  to the horizontal, where  $\sin \theta = \frac{1}{30}$ . The resistance to the motion of the truck is again modelled as a constant force of magnitude  $R$  newtons. The engine of the truck is again working at a constant rate of  $P$  watts.

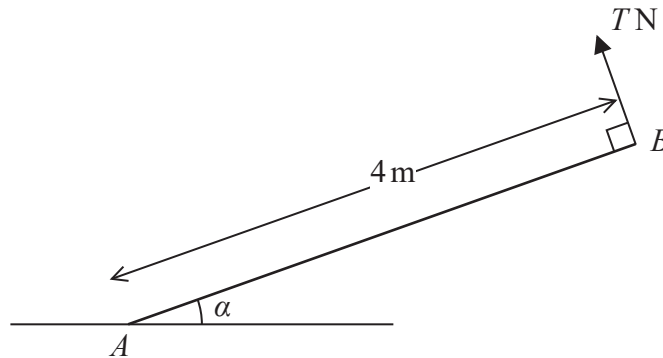
At the instant when the speed of the truck is  $12\text{ m s}^{-1}$ , the acceleration of the truck is  $0.4\text{ m s}^{-2}$

Find the value of  $R$ .

(8)



5.



**Figure 2**

A uniform rod  $AB$  has length 4 m and weight 50 N.

The rod has its end  $A$  on rough horizontal ground. The rod is held in equilibrium at an angle  $\alpha$  to the ground by a light inextensible cable attached to the rod at  $B$ , as shown in Figure 2. The cable and the rod lie in the same vertical plane and the cable is perpendicular to the rod. The tension in the cable is  $T$  newtons.

Given that  $\sin \alpha = \frac{3}{5}$

(a) show that  $T = 20$

**(3)**

Given also that the rod is in limiting equilibrium,

(b) find the value of the coefficient of friction between the rod and the ground.

**(6)**



6. Two particles,  $P$  and  $Q$ , are moving in opposite directions along the same straight line on a smooth horizontal surface so that the particles collide directly.

The mass of  $P$  is  $km$  and the mass of  $Q$  is  $m$ .

Immediately before the collision, the speed of  $P$  is  $x$  and the speed of  $Q$  is  $y$ .

Immediately after the collision,  $P$  and  $Q$  are moving in the same direction, the speed of  $P$  is  $v$  and the speed of  $Q$  is  $2v$ .

The coefficient of restitution between  $P$  and  $Q$  is  $\frac{1}{5}$

The magnitude of the impulse received by  $Q$  in the collision is  $5mv$

- (a) Find (i)  $y$  in terms of  $v$   
(ii)  $x$  in terms of  $v$   
(iii) the value of  $k$

(9)

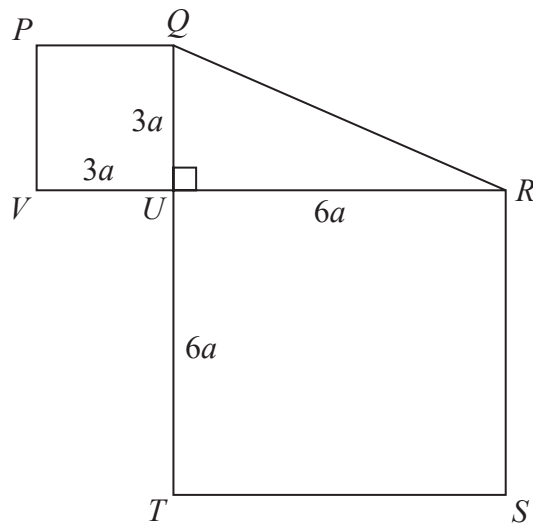
- (b) Find, in terms of  $m$  and  $v$ , the total kinetic energy lost in the collision between  $P$  and  $Q$ .

(3)

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7.



**Figure 3**

The template shown in Figure 3 is formed by joining together three separate laminas. All three laminas lie in the same plane.

- $PQUV$  is a uniform square lamina with sides of length  $3a$
- $URST$  is a uniform square lamina with sides of length  $6a$
- $QRU$  is a uniform triangular lamina with  $UQ = 3a$ ,  $UR = 6a$  and angle  $QUR = 90^\circ$

The mass per unit area of  $PQUV$  is  $k$ , where  $k$  is a constant.

The mass per unit area of  $URST$  is  $k$ .

The mass per unit area of  $QRU$  is  $2k$ .

The distance of the centre of mass of the template from  $QT$  is  $d$ .

(a) Show that  $d = \frac{29}{14}a$

(5)

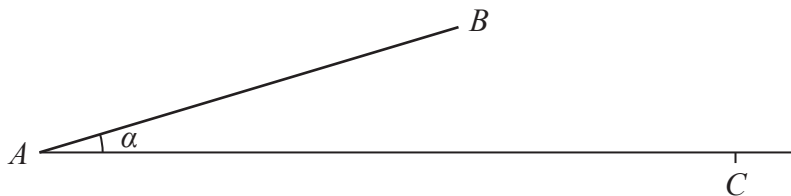
The template is freely suspended from the point  $Q$  and hangs in equilibrium with  $QR$  at  $\theta^\circ$  to the downward vertical.

(b) Find the value of  $\theta$

(7)



8.



**Figure 4**

Figure 4 shows a rough ramp fixed to horizontal ground.

The ramp is inclined at angle  $\alpha$  to the ground, where  $\tan \alpha = \frac{1}{6}$

The point  $A$  is on the ground at the bottom of the ramp.

The point  $B$  is at the top of the ramp.

The line  $AB$  is a line of greatest slope of the ramp and  $AB = 4$  m.

A particle  $P$  of mass  $3$  kg is projected with speed  $U$  m s<sup>-1</sup> from  $A$  directly towards  $B$ .

The coefficient of friction between the particle and the ramp is  $\frac{3}{4}$

(a) Find the work done against friction as  $P$  moves from  $A$  to  $B$ .

**(4)**

Given that at the instant  $P$  reaches the point  $B$ , the speed of  $P$  is  $5$  m s<sup>-1</sup>

(b) use the work-energy principle to find the value of  $U$ .

**(4)**

The particle leaves the ramp at  $B$ , and moves freely under gravity until it hits the ground at the point  $C$ .

(c) Find the horizontal distance from  $B$  to  $C$ .

**(6)**